# IMPROVING GRIP ON A PROSTHETIC HAND

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## Introduction

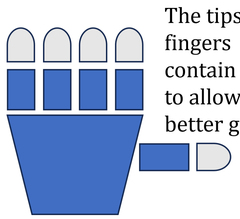
**The Problem:** Lack of grip strength and overall gripping ability amongst various prosthetic hands.
**Our solution:** Engineer adaptions into the e-NABLE hand prosthetic to allow additon of grippier/tackier material where the fingertips and palm are.

## Phases of Research:

1. Research various solutions to improving grip.
2. Look into various 3D printed hands that allow **modifications**, where we selected the **e-NABLE** hand [1].
3. Focus on **improving grip** specifically on **holding awkward/round objects** (where the prosthetic field is lacking the most solutions).

## Design:

1 We started our design with looking into different ways we could improve the grip, such as adding silicone, polymer, or using filament in different areas, since the original PLA option is very slippery and PLA is known to have a low coefficient of friction.



The tips of the fingers contain silicon to allow for better grip.

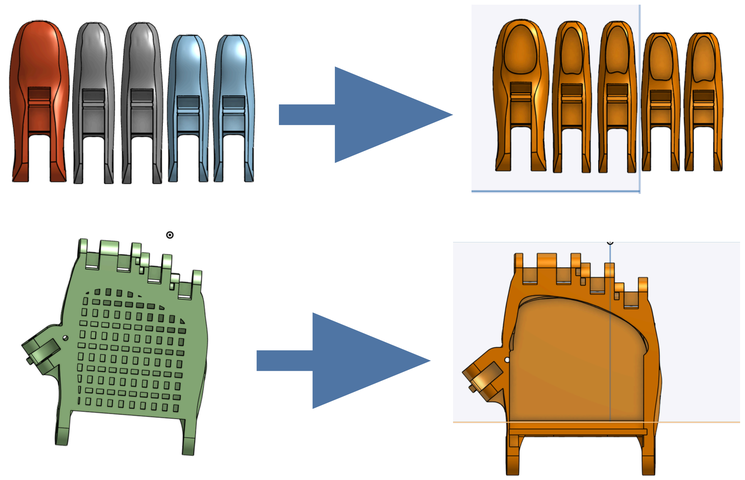
2 **Our design criteria:**
we narrowed down our design to modify the e-NABLE hand with cutouts filled with polymer.

| | Cost (2) | Grip strength (3) | Hold awkwardly shaped objects (4) | Total |
|---|---|---|---|---|
| Adding fingerprints to tips. | 1<br>1*2=2 | 1<br>1*3=3 | 1<br>1*4=4 | 9 |
| **Grip on fingertips and palm** | 2<br>2*2=4 | 2<br>2*3=6 | 2<br>2*4=8 | 18 |

## Design Cont.

3 We took the current e-NABLE hand, and used Onshape to modify the original hand design. Using **spline**, **extrude**, **remove** functions to create divots in the fingers and palm.

**Before** **After**



**Our Modifications:**

- Adding divots where the fingerprints used to be, and a large divot on the palm for a tacky material to fill.
  - This replicated grip in human fingertips and palms.

We used a petroleum-based EVA which allowed for a firm and tacky feel on the prosthetic.
We chose to use EVA polymer because it is more cost effective, flexible and overall sturdier when compared to other fill options like liquid silicone.

4 We 3D printed the unmodified hand and the modified hand in order to compare the two.



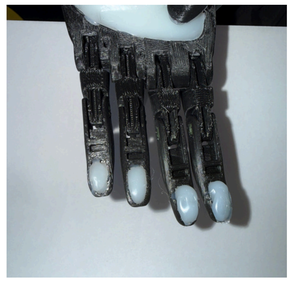
## Design Cont.

5 Finally, using a hot glue gun, we filled the divots with EVA polymer, creating grip on the e-NABLE hand.

When filling the divots with the EVA polymer we left it to dry unevenly. This was to imitate the uneven texture of a human palm. We also made sure the fingertips dried in a naturally smooth, round shape.

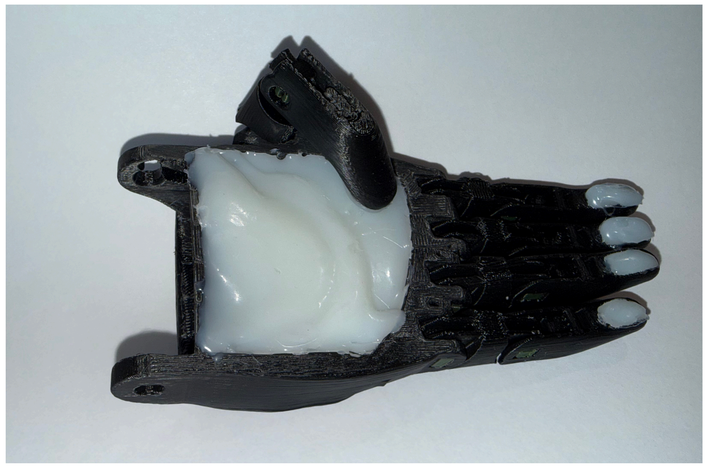


Before



After

## Final Product



Our completed 3D printed modified e-NABLE hand with improved grip capabilities.

## Future Implementation

1. Survey possible users for feedback on the product.
2. Test grip strength capacity with various experiments or machinery.
3. Find a cost similar alternative to EVA polymer for the grip.
4. Work with engineers of different fields for inputs on how to improve the design and make it more functional

[1] e-Nable, “Enabling The Future,” Enabling The Future, 2023. https://enablingthefuture.org/